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A Novel Machine Learning Approach to Predict Gas Content in Unconventional CBM Reservoir from Geophysical Logs, a Supplementary of the Desorption Method

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Abstract

Gas content is a key parameter for CBM development. Traditional GC measurement via core desorption is time-consuming, costly and limited. This study aims to develop a machine learning (ML)-based model to predict gas content (GC) in coalbed methane (CBM) reservoirs using geophysical well logs. The ML approach offers a faster, cost-effective addition to estimate GC in development wells where core is not taken.

Gas content and geophysical log data (Gamma, Resistivity, Density) of some exploratory core wells from Eastern part of Raniganj basin in India, were used for the study. Core data from those wells were used for training and testing purpose of the model. Four ML models – Random-Forest, Decision-Tree, SVM, KNN-Regressor; were used for training purpose. To improve model performance, hyper parameters were being tuned for all models. Cross validation was done in each step of hyper parameter tunning to check accuracy. Feature selection and cross correlation performed for getting most relevant geophysical log.

Good accuracy achieved for all four models when used on trained core wells. However, in test core wells, highest accuracy has been achieved using Random-Forest model where other models achieved moderate accuracy. Gas content predicted by the Random-Forest model in test core wells matched closely with lab measured data. Some of those test core wells achieved accuracy more than 90% match and rest had a match range of 75-90%. Random-Forest model is being used for further prediction in development wells. GC predicted in development wells using Random-Forest model, giving significant match with nearest exploratory core well, showing a good indication of model performance in development wells. The Resistivity log was found to be the most relevant geo-physical log for gas content prediction.

This ML approach enables us to go beyond traditional approach and predict GC in development wells which generally have the required geophysical logs. Due to large number of development wells this method gives a significant control on GC data. This enables operator to identify the heterogeneity and improve gas production forecast ability. Additionally, this help us in well production performance analysis and optimize field development planning. This can be implemented in other fields by tunning some parameters of the model.

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Introduction

The global persuasion of net zero emission has led the world through a transition in last decade. In this journey of slow and steady shift towards sustainable clean energy resources; Coal Bed Methane (CBM) has a key role to play as one of the critical transitional fluid. CBM production and potential is largely dependent upon three major criterions – Coal seam thickness, Gas content and Permeability. It is absolutely crucial to measure the gas content for Resource estimation for any CBM field. In general, CBM development is done in stepwise manner by building confidence through establishing productivity. The average well count to develop a CBM field is often significantly higher than any conventional field of similar area. With each well contributing towards new understanding CBM field development plan goes through multiple modifications and optimizations. Leveraging this vast data created across the exploration and development phase combining with the advanced data analytics and machine learning, companies can gain deeper insights into the true potential, predict reservoir behavior and optimize the drilling targets.

Gas content and subsequent resource estimation becomes a key decision point for such optimization throughout the development of CBM field. Globally most practiced method for gas content estimation is to collect core and perform desorption study. While core desorption is time taking and cost intensive process, accordingly it is common to drill one core well for 2 – 2.5 sqkm, which is rather inadequate to address the natural heterogeneity of coal seams. To reduce this uncertainty, the researchers tried to estimate gas content of coal seams based on the geophysical logs. As geophysical logs are recorded in most of the wells a vast data Machine Learning (ML) model can predict gas content of the wells using geophysical logs in development wells. This paper proposes Machine learning driven solution for log based gas content estimation using geo-physical logs. This study was conducted on the data from CBM block in the Eastern most part of the Raniganj Basin, operated by Essar Oil and Gas Exploration and Production Limited (EOGEPL). With ~500 wells including ~50 exploration and ~ 450 production wells drilled in the area, EOGEPL has good data control to build such robust model and effectively utilize the same for further development planning. While gas content and subsequent resource estimation a key decision point for strategic development of CBM field; the Data-driven decision-making enables fast, accurate resource estimation and efficient planning. This approach also helps to minimize exploration and development costs, reduce uncertainty and maximize returns on investment, ultimately leading to more informed and effective decision-making and efficient development planning.

Study Area

The Raniganj Coalfield, one of India's oldest coal-producing regions, forms the easternmost extension of the Damodar Valley Coalfields. This block lies within the Durgapur Depression, situated in the district of Bardhaman, West Bengal. The studied block lies in the easternmost part of the block. The Raniganj Formation in the block is consistent with 12 to 14 individual coal seams with thickness varies from 0.5m to 10m. Based on depositional setting and regional correlation the coals seams are grouped in six regional group of coal seams, i.e. Seam-1 (the bottom most) to Seam-6 (top most) group of coal seams.

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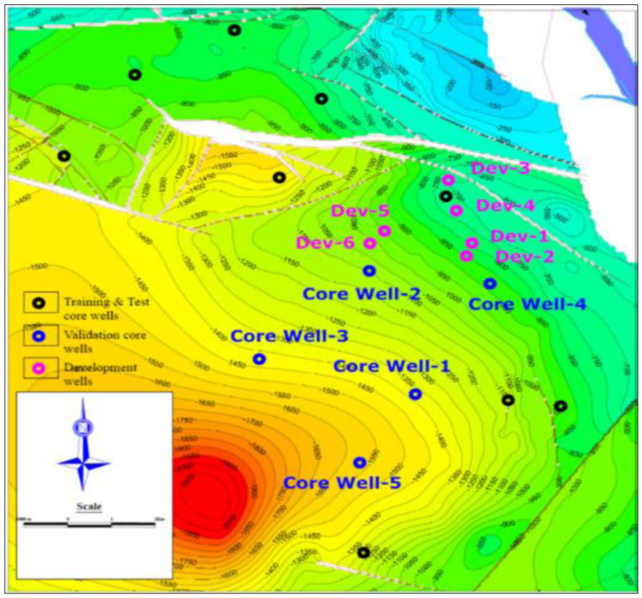

Figure 1—*Study Area showing location of different training, validation and development core wells*

Splitting merging of the seams are common across the block. The bottom most Seam – 1 and the one above Seam – 2 are merged i.e. present as a single seam in most part of the study area and splitted (i.e. present as 2 individual seams that cumulatively represents almost similar thickness as the merged one) in rest of the part. As the thickness of Seam - 1 is negligible in the parts where the seams are split; for this study in split sections Seam – 2 is taken under analysis and where the seam is present as merged – it is identified as Seam-2. Cumulative coal thickness varies significantly across the block. The average cumulative thickness of the fairways is 30-50 m. Coal thickness generally increases at center of the block which goes up to 50-60 m. The northern and southern part of the block has relative less coal thickness which increase towards the center (from<10 m to 50-60 m). A similar trend is observed in coal thickness from West to East.

Table 1—Avg Seam Wise Properties in the Study Area summarized below

Raniganj Coal Seams	Depth,	Initial Gas Content,
	(m)	(cc/gm)
Seam-6	350 - 900	4 - 7
Seam-5	400 - 980	6 - 8
Seam-4i	450 - 1050	6 - 8
Seam-4ii	500 - 1100	6 - 10
Seam-3	550 - 1200	8 - 11
Seam-2	600 - 1250	7 - 10

Data Preparation

Conventional Method of Gas estimation

The traditional desorption technique involves retrieving core samples from coal seams and quantifying gas released under controlled laboratory conditions. The procedure includes measuring (i) lost gas (Q1, estimated volume lost before sealing) (ii) Measurable desorbed gas (Q2, released after coring), and (iii) residual gas (Q3, remaining after crushing the sample).

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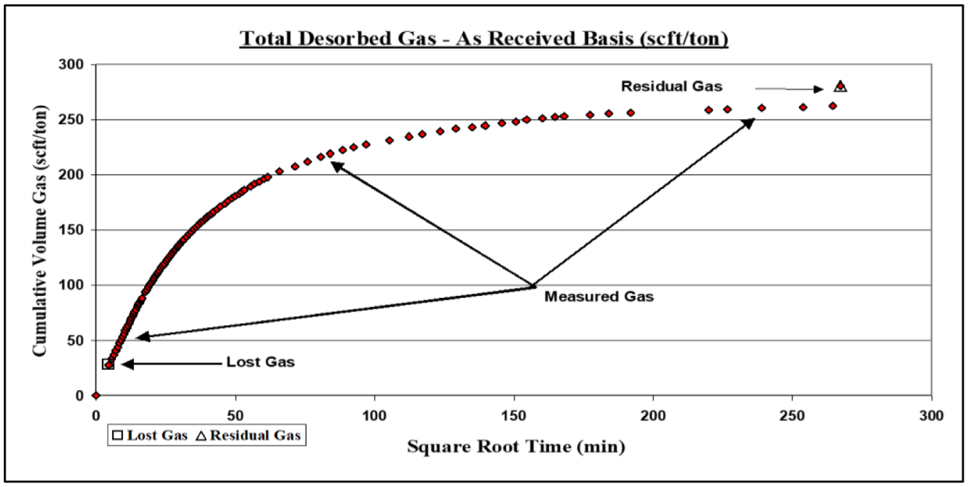

Figure 2—*Total Desorbed Gas (scf/ton)*

Lost Gas component (Q1). The gas released from a coal core sample prior to being placed in a canister for desorbed gas measurement is termed 'lost gas.' This value is typically higher for deeper coal seams and traditional coring methods. The calculation of lost gas is performed using a graphical approach, based on the principle that, during the initial desorption period, the gas volume released is proportional to the square root of the time elapsed, as shown in Figure 3 below.

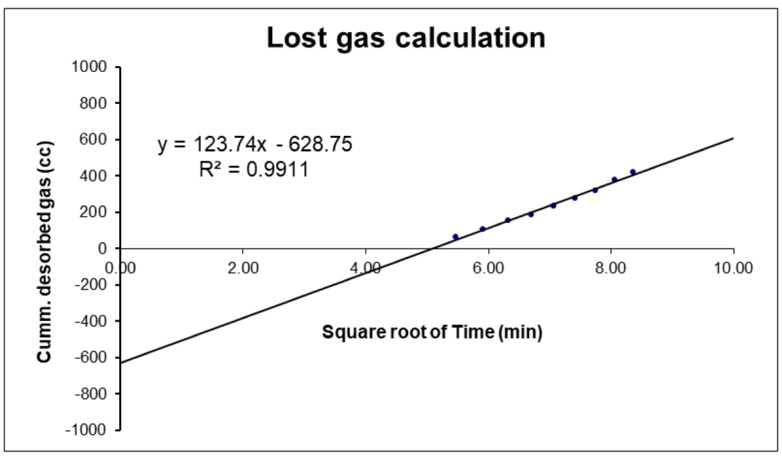

Figure 3—Lost Gas Calculation from graphical method

Measurable desorbed gas(Q2). Q_2 is the measurable gas desorbed from the non-pulverized coal sample while it remains sealed in a desorption canister at atmospheric pressure and ambient temperature. The samples are immediately placed in desorption canisters, weighed, and then transferred to a pre-heated water bath (maintained at reservoir temperature for at least one hour before retrieving the core) for gas content testing. Initial gas release is measured frequently (e.g. every 5 minutes for the first hour), then at longer intervals (e.g. every 10–15 minutes for a few hours), followed by daily measurements until gas evolution falls below a threshold (e.g. ≈ 0.05 mL/g/day). Data collection continues until desorption stabilizes or no further gas is released over a five-day period. The coal is then carefully extracted, split in half, with one portion used for measuring residual gas content and performing proximate analysis.

Residual Gas (Q3). Residual gas is the portion of total gas which is not released during desorption period. A small part of the coal core sample tested for desorption is taken out, weighed and put into an air tight steel vessel (Rod mill), which contains several brass rods to grind the core sample as shown in Fig-4. The

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steel vessel is initially purged with Nitrogen to avoid absorption of oxygen in the air present in the vessel, by crushed coal. The coal sample is crushed to fine powder below 200 BSS size in about 40 minutes.

Figure 4—*Desorption Method 4(a) & 4(b) Examples of Water Bath maintained at reservoir temperature; 4(c) Manometric gas volume measurement instrument*

The total volume of gas is obtained by adding Q1, Q2 and Q3.

$$\text{Gas content} : Q\ (cc/gm) = (Q1 + Q2 + Q3)/W$$
$$W = \text{Weight of the coal core sample.}$$

The total gas content (Q) is calculated as the sum of these three components. While this method provides accurate, direct measurements and is considered an industry standard, it is time-consuming, costly, and limited to a small number of cored wells, restricting its practical application for large-scale studies. This method for coal gas content estimation is widely recognized for its accuracy but time consuming and often costly. This is primarily due to the need for specialized coring operations, immediate sample handling, long laboratory desorption periods, and detailed measurements of lost and residual gas.

Geophysical Log response of coal seams.

- Gamma Ray Log

  Gamma ray logs measure the natural radioactivity of formations, primarily detecting radioactive isotopes of uranium, thorium, and potassium. Higher gamma readings typically indicate shales or clay-rich rocks, which contain more radioactive elements. Coal has very low natural radioactivity,

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so it shows low gamma-ray counts. This contrast with surrounding shaly formations helps in identifying coal seams clearly. In case of shaly-coal the gamma count lies in between coal and shale.

- Density Log

  Density logs measure the electron density of formations, which correlates with bulk density. Higher density rocks (like sandstones and carbonates) produce higher readings, while less dense rocks show lower values. Coal is less dense due to its higher organic matter content and porous structure, appearing as a pronounced low-density on the density log.

- Resistivity Log

  Resistivity logs measure how strongly a formation resists the flow of electrical current. Fluids in pores (like water or hydrocarbons) affect resistivity values - water-saturated rocks have lower resistivity, while hydrocarbon or gas-bearing rocks have higher resistivity. Coal typically exhibits higher resistivity compared to water-saturated shales and sandstone.

- Acoustic (Sonic) Log

  Acoustic logs measure the travel time of sound waves through rock formations. Denser, more compact rocks transmit sound faster (lower transit time), while softer, porous rocks slow sound wave down (higher transit time). Due to its softness and higher porosity, coal exhibits slower sound velocity, resulting in higher acoustic transit times.

- Caliper log

  Caliper log records the borehole diameter with depth, helping identify wellbore enlargements or contractions. It is crucial for assessing borehole stability and ensuring the reliability of other log measurements, especially in zones prone to washouts or formation collapse.

By integrating these log responses, it is possible to identify the coal seams and select the target seam for completion based on their quality, as represented in Fig -5.

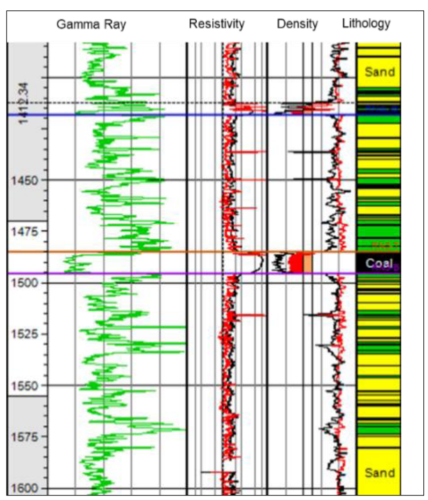

Figure 5—*Coal Seam Identification from Geophysical Logs*

Machine learning (ML) methods combined with geophysical logging technology have enabled advanced, in situ estimation of gas content in coalbed methane (CBM) reservoirs. Basic geophysical logs such as density (DEN), resistivity (Res), and gamma ray (GR) have been used as input data in ML models to predict gas content. Machine Learning-Enabled Geophysical Logging as an Alternative to Conventional Desorption for Gas Content Estimation (Zhang et al., 2025), is a cost effective and time saving solution, especially suitable for optimizing development strategy at beginning or by mid of a running campaign.

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Data Cleaning and Quality Control. Data cleaning is one of the crucial part of ML model training. Dataset quality should be good enough to train the model for extracting relevant features and properly predict the output. Model may be irrelevant and ineffective if trained on bad dataset.

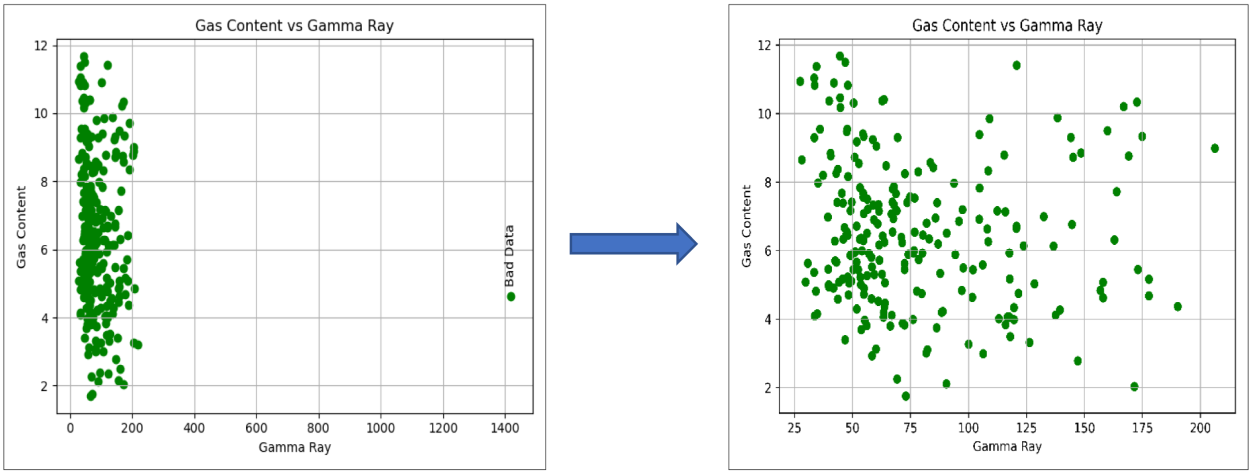

Figure 6—*Gamma Ray distribution 6(a) with Bad Data; 6(b) without Bad Data*

Also, in the Gas content with Gamma Ray plot, bad data has been identified and removed from the dataset. Cleaned data has been used for further works.

For this study the data set was prepared based on correlation of log properties and subsequent physical properties as identified in core wells and subsequent quality control was applied based on the geophysical logs recorded. After quality control of the dataset, outliers were discard by using cutoff filters.

As the density of coal cutoff taken to be 2 gm/cc, density greater than 2gm/cc had been eliminated. Also, density less than 1 gm/cc had been eliminated as bad dataset as such data often a result of well bore condition rather than the true rock property. The distribution of density before and after the quality control can be seen in Fig-7.

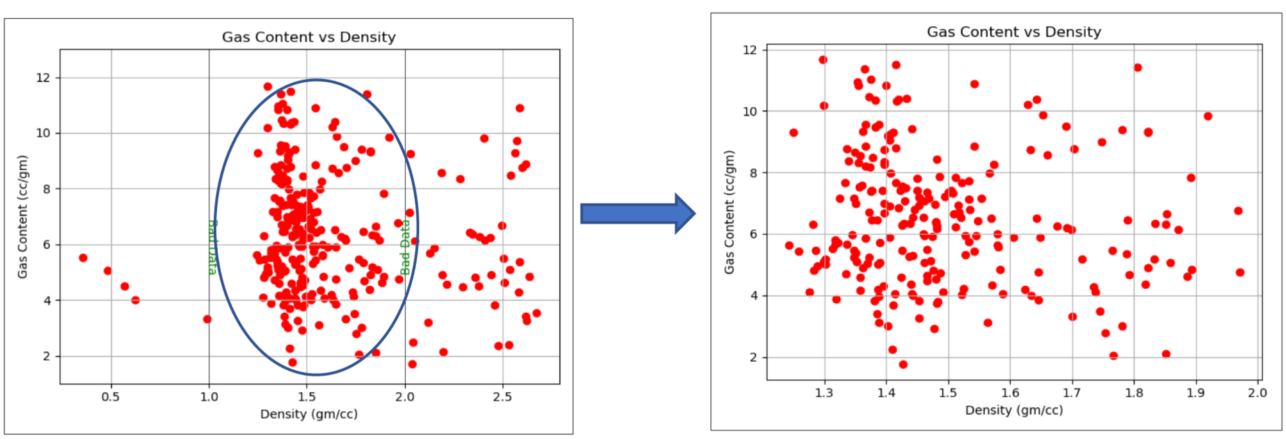

Figure 7—*Gas Content with Density Distribution 7(a) Before Density Correction; 7(b) After Density Correction*

Exploratory Data Analysis

Core wells gas desorption dataset is the only source for core well gas content measurement. Along with that geophysical log response of core wells have been used. Around 300 data points from 10 core wells in Raniganj Basin distributed across 180 sq-km area have been used for this study. Samples have been collected seam-wise, as different seams have different properties. Raw GC data have been collected from

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core sample after coring operation. The geophysical log response of the same wells has been collected. Correlation of the seam wise lab and geophysical data exhibits that both the log response and GC varies seam wise as well as depth wise. While the log response generally remains similar for specific seam but the gas content varies both laterally and vertically.

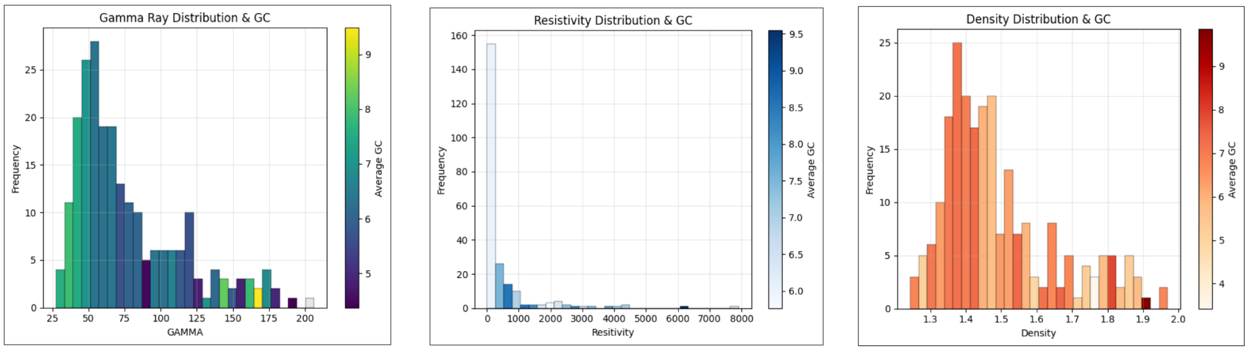

Figure 8—*Histogram Distribution of Geophysical Logs 8(a)Gamma Ray Distribution; 8(b) Resistivity Distribution; 8(c) Density Distribution*

The statistical analysis of geophysical logs reveals distinct distribution patterns across the dataset. Gamma Ray (GR) values range from 28 to 200 API, with a mean of 77 API, indicating that most formations are relatively clean, as 75% of values lie below 97.7 API. Resistivity exhibits high variability (mean: 461.75 ohm-m; std: 1000.77) with a pronounced right-skew, reflecting the presence of high-resistivity coal seams. Density values are more uniformly distributed (mean: 1.495 gm/cc), with 50% of the data falling between 1.381 and 1.567 gm/cc. Distribution can be seen in Table-2.

Table 2—Geophysical Logs Statistical Distribution

Geophysical Log	Mean	Std	Min	Max	25%	50%	75%
Gamma Ray	77.934716	38.09836	27.45	206.2	50.38	64	97.7
Resistivity	461.746725	1000.771	20	7910	30	80	404
Density	1.495214	0.161564	1.243	1.97	1.381	1.45	1.567

Statistical Correlation. Pearson Correlation: The Pearson correlation coefficient is calculated using the covariance of the variables, normalized by dividing by the product of their standard deviations. This normalization makes the result independent of the scale of the variables.

Kendall Correlation: Kendall's tau measures the association between two variables by assessing the monotonic relationship—that is, whether the variables tend to increase or decrease together - rather than focusing solely on linearity like Pearson's r.

Spearman's Rank Correlation: Spearman's rho is a non-parametric statistic used to evaluate the monotonic relationship between two variables. It measures the strength and direction of the association based on the rank order of the data, rather than their actual values.

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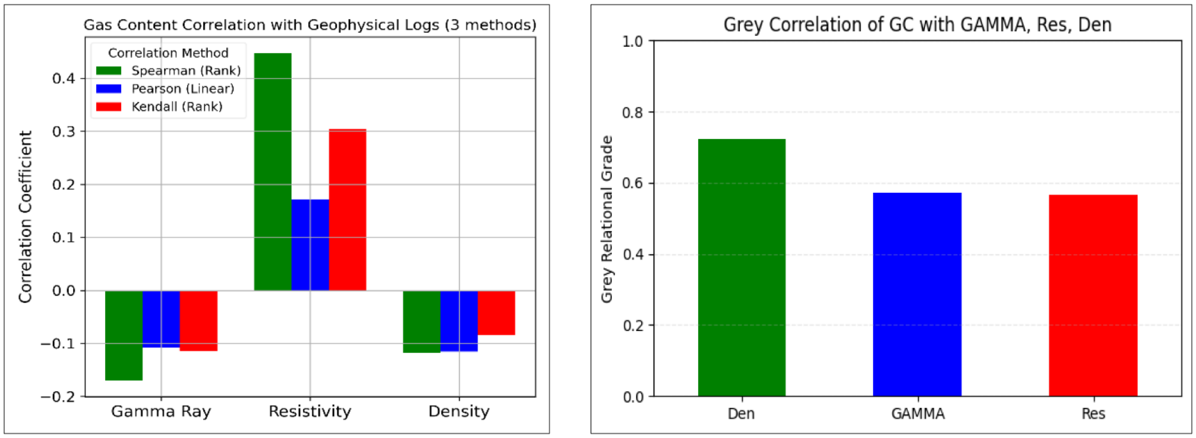

Figure 9—*Statistical Correlation of Gas content with Gamma, Resistivity and Density 9(a) Pearson, Spearman and Kendall methods; 9(b) Grey Correlation*

Grey correlation:

Grey Relation is a measure of similarity or closeness between two sequences (or sets of data), often used to evaluate the relationship between multiple factors in a system. Correlation analysis between GC and the input parameters reveals varying strengths and types of relationships. GAMMA exhibits a consistently weak negative correlation across all methods - Spearman (- 0.17), Pearson (- 0.10), and Kendall (- 0.11) - indicating a minor inverse association. Resistivity, on the other hand, shows a positive correlation, particularly strong in the non-parametric methods - Spearman (0.42) and Kendall (0.28) - suggesting a potential non-linear relationship. Pearson's lower value (0.12) supports this interpretation. Among all parameters, resistivity appears to be the most positively correlated with GC based on traditional statistical measures. Density demonstrates a mild negative correlation (~ -0.1) across all three methods, implying a weak inverse influence. However, Grey Relational Analysis offers a different perspective. The Grey Relational Grade (GRG) for density is 0.72, indicating the strongest relationship with GC under this method. GAMMA and resistivity both show a moderate GRG of 0.56, suggesting a moderate influence. These results highlight that while resistivity dominates in traditional correlation metrics, density has the highest influence in the grey system context, pointing toward possible non-linear or complex interactions.

Machine Learning (ML) Process and Methods

Gas content is directly linked with coal maturity and property. Geophysical logs represent the natural rock properties. In case of development wells geophysical logs are the key tool to identify not only the coal seams (J et al., 2017), but also to optimize the completion target and hydro-fracturing strategy. In this paper researchers are proposing method to extend the log based method from coal seam and quality estimation to quantification of gas content.

Feature Selection

Feature selection is a fundamental process in machine learning aimed to identify the most relevant input variables for predictive modelling. This process not only enhances model performance but also improves computational efficiency and interpretability (Guyon & Elisseeff, 2003).

For feature selection, basic geophysical well logs - Gamma Ray, Density, and Resistivity - were utilized to predict Gas Content (GC).

It is rather common observation for coal seams to have better maturity with depth and subsequently having better gas content leading to a good correlation between depth and gas content. But incorporating depth directly as a predictive feature proved challenging in this study as there are multiple seams encountered across the depth having varied coal property and gas content. The geological heterogeneity of the study

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area, with coal seams distributed across a wide depth range (approximately 400–1600 m), makes depth a less reliable standalone predictor. In similar depth intervals, GC values can vary substantially between different seams, as illustrated in Fig-10 & Fig-11, highlighting the limitations of using depth alone. To address this, rather than treating depth as a continuous variable, the model incorporates seam continuity as a categorical feature. The region contains multiple major seams - seam_2, seam_3, seam_4, seam_4i, seam_4ii, and seam_6 - each exhibiting distinct yet internally consistent characteristics. This seam-based approach captures geological variations more effectively than depth only. Additionally, the nearest core well can be included as optional information during model training, potentially enhancing prediction accuracy by leveraging lateral continuity.

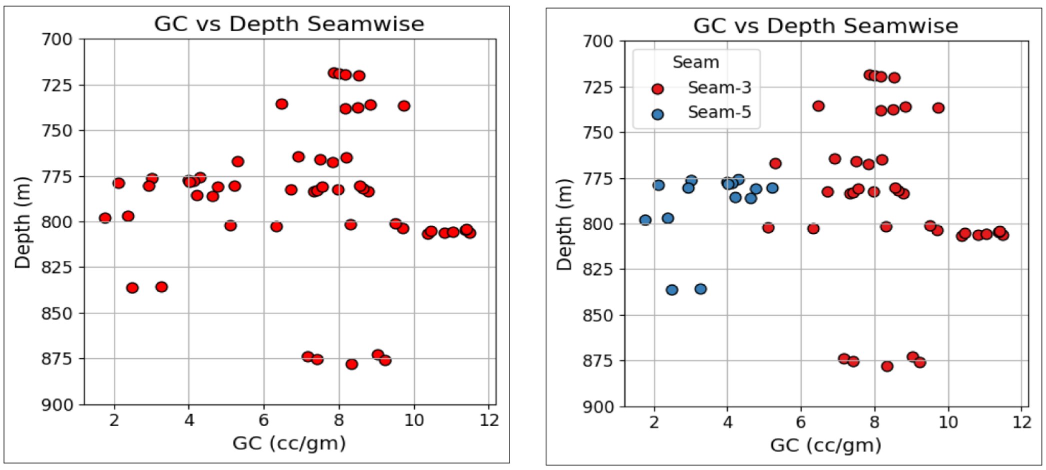

Figure 10—*(a) Gas Content with Depth; (b) Seam-wise (Seam-3 and Seam-5) Gas Content With Depth*

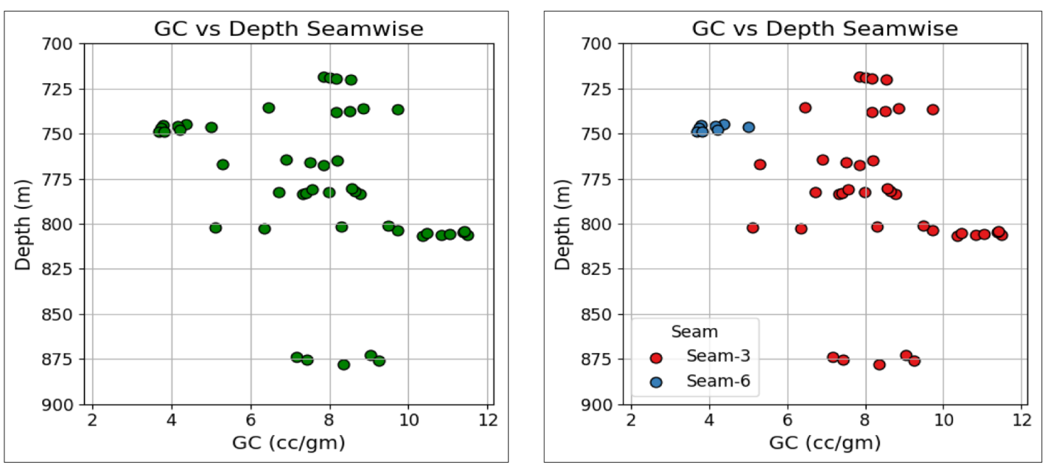

Figure 11—*(a) Gas Content with Depth; (b) Seam-wise (Seam-3 and Seam-6) Gas Content With Depth*

Model Building

To predict Gas Content (GC)**,** five different machine learning algorithms were employed: Decision Tree**,** Random Forest**,** Support Vector Regression (SVR)**,** Multilayer Perceptron Regressor (MLP), and XGB Regression. These algorithms were selected based on their distinct characteristics and proven effectiveness in handling various types of regression problems. Each model offers unique strengths - such as interpretability, non-linear mapping capabilities, and robustness to overfitting - making them suitable for exploring the complex, non-linear relationships observed in the dataset.

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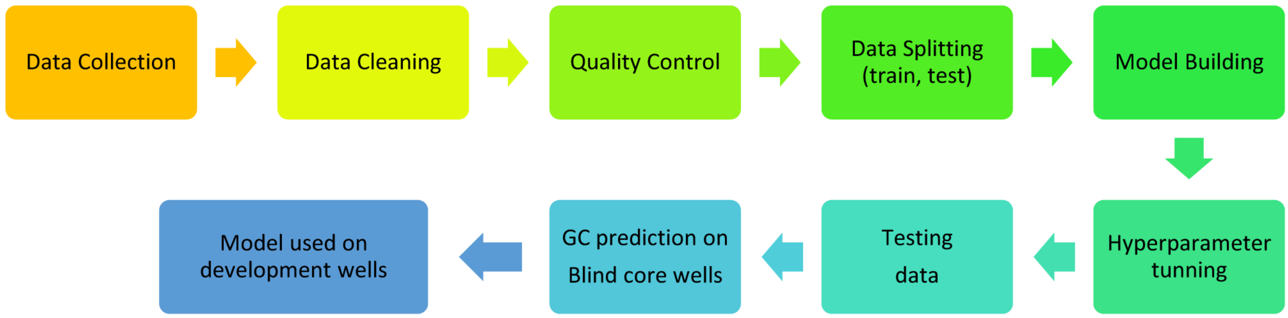

Data Transformation

The cleaned dataset consists of numerical features - including Gamma Ray, Density, and Resistivity - as well as categorical features, such as seam identifiers and information about the nearest core wells. To enable the use of categorical variables in machine learning models, appropriate encoding techniques are required to transform these categorical inputs into a numerical format. This step ensures that the models can interpret and learn from all relevant features effectively, maintaining the integrity and predictive value of both numerical and categorical data during training.

Data Splitting

In machine learning, the dataset is often split into two subsets: a training set and a test set, typically using an 80-20 or 70-30 ratio. The training dataset (80% of dataset) was being used for training the model, enabling it to capture patterns and complex relation in between the data. The test dataset (the remaining 20%) was kept separate and used for the evaluation of the performance of the model following that it has been trained. This test set serves as a proxy for new, data that is not seen, helping to assess the generalization ability of the model and ensuring it is not overfitting to the training data (Baldwin, C., et al.,2021). By using this method, we can better understand how the model will perform when applied to real-world, unseen data, making it a critical phase for machine learning workflow.

Applied ML Models

Decision Tree Regressor. The Decision Tree (DT) model is a simple and interpretable machine learning algorithm widely used for both classification and regression tasks. It works by iteratively partitioning the dataset based on feature values to minimize variance (in regression) or impurity (in classification) within the resulting subsets. DT is particularly beneficial since it can effortlessly manage both numerical and categorical data without requiring complex pre-processing or transformations (Loh, W.-Y., 2011). They also excel at capturing non-linear relationships between features. Moreover, their visual representation as tree-like structures makes them easy to understand and interpret. However, decision trees often tend to overfit if they are too deep or complex, and their performance can be unstable with small variations in the data.

Random-forest Regressor. A Random-forest Regressor is a combination of multiple decision trees that use to improve prediction accuracy alongside reducing overfitting. Training of each tree has been done on a random subset of data and selects subset consists of features randomly for each split, ensuring diversity among the trees. Final prediction evaluate the average of the individual tree predictions, which helps smooth out errors and variance. This technique widely used for regression tasks, as it leverages the strengths of multiple trees to build a more reliable and high-performing model (Breiman, 2001).

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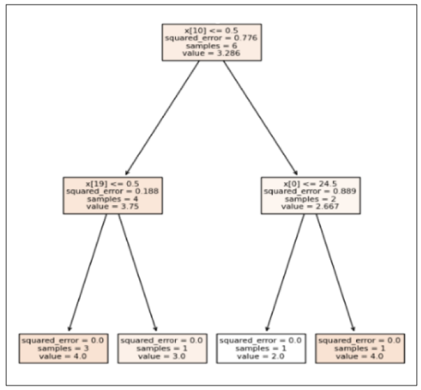

Figure 12—*Decision Tree Regression Structure*

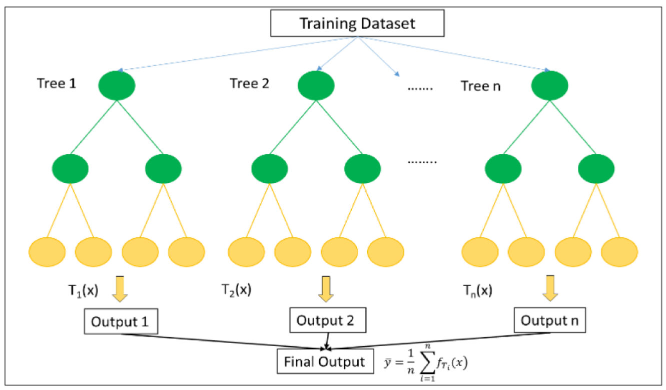

Figure 13—Random Forest Regression Structure

SVR (Support Vector Regression). One kind of Support Vector Machine (SVM) utilized for regression tasks is called Support Vector Regression (SVR). SVR attempts to fit the best line (or hyperplane in higher dimensions) from the actual data points within a margin of tolerance (epsilon, ε), in contrast to typical regression models that seek to minimize the error between predicted and actual values. According to Smola and Schölkopf (2004), points falling within this range are regarded as acceptable and do not affect the outcome.

MLP-Regression. MLP Regression uses a neural network called a Multilayer Perceptron to model complex, non-linear relationships between input features and continuous target values. It consists of an input layer, one or more hidden layers with non-linear activation functions, and an output layer that produces the final regression prediction. Learning of the network has been done by adjusting weights and biases using backpropagation, minimizing loss function as an example mean squared error (Ouadfeul & Aliouane, 2015). MLP Regression is flexible and powerful, capable of capturing intricate data patterns.

XGBoost Regression. XGBoost Regressor is a very efficient algorithm in machine learning based on gradient boosting, designed for high-performance regression tasks. An ensemble of decision trees has been built sequentially, the prediction errors of its predecessors has been corrected by each tree. Key features such as regularization, parallel computation, and efficient handling of missing data contribute to its robustness and scalability. By optimizing a loss function - commonly Mean Squared Error - XGBoost achieves high accuracy and generalization, making it well-suited for structured data regression problems (Chen & Guestrin, 2016).

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Model Training

For model training, we need prepare dataset. Training dataset (80% of dataset) used to train the model to learn patterns or features from the data. Model hyper parameters are being tunned to make the model learn better from the dataset. Hyper parameters tunning may lead to overfitting the model. Checking model score with respect to both training data and testing data, helps us to stay alert. If the model score is good for training data but poor for testing data. Grid search algorithm has been user to tune hyper parameters. In this method multiple sets of hyper parameters choosed. Then iteration has been done to fit the model using different combinations of hyper parameters (Bishop, 2006). In each iteration, training and testing score have been calculated. Hyper parameters with best training & testing score have been choosen for the model training.

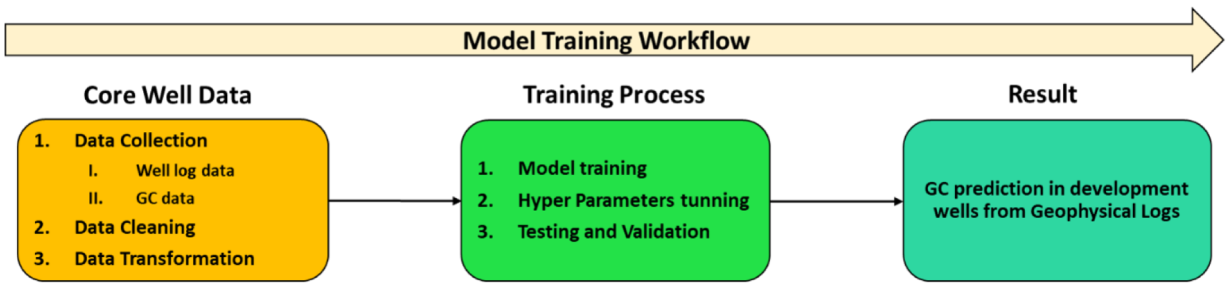

Figure 14—Model Training Workflow

During model training, particularly for regression tasks, performance evaluation is critical to assess the predictive capability of the model. Two widely used metrics for this purpose are the Mean Squared Error (MSE) and the Coefficient of Determination (R^2 score).

Mean Squared Error (MSE): MSE quantifies the average of the squared errors, which is the average squared difference between the actual and predicted values. Mainly focus on larger error, minimize the weightage of small error due to squaring. Lower MSE means model is good for prediction.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y}_i)^2$$

Where, y_i denotes actual value. $\bar{y}_i$ signifies predicted value. n is the no of data points.

R^2 Score (Coefficient of Determination): R^2 indicates the fraction of variability in the dependent variable that can be predicted from the independent variables. Ranges from negative infinity to 1. R2 closer to 1 means better fit, where 0 means the model predicts nothing good than the mean; negative values mean worse than the mean.

$$\mathrm{R2} = \frac{\sum(y_i - \bar{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Where, y_i represents actual value, $\bar{y}_i$ denotes predicted value and $\bar{y}$ signifies the mean of actual values. Models are being evaluated based on R2 score. Among the models, RF and DT regression model achieve highest accuracy and best R2 score.

Results

The performance of the regression models has been evaluated on both the training and testing datasets using standard metrics, including the R^2 score, Mean Squared Error (MSE), and Mean Absolute Error Percentage (MAPE). This evaluation provides a comprehensive understanding of each model's accuracy, generalization ability, and error behaviour across different data splits.

Training and Testing results comparison of different models given in Table-3.

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Table 3—Comparison of ML Score in Training and Testing Data

Model	Training Data			Testing Data		
	R^2 Score	MSE	Mean Absolute Error (%)	R^2 Score	MSE	Mean Absolute Error (%)
Random Forest	0.92	0.37	8.7	0.78	0.80	10.8
XGB Regression	0.88	0.559	10.5	0.77	0.86	12
Decision Tree	0.81	0.77	12.7	0.77	0.85	10.8
MLP Regression	0.70	1.38	16	0.70	1.1	11
SVR	0.72	1.19	14	0.65	2.17	21
KNN-Regressor	0.40	4.09	23	0.23	7.42	35

This section evaluates, the performance of five regression models - Random Forest, Decision Tree, Support Vector Regression (SVR), Multi-Layer Perceptron (MLP) Regression, and XG-Boost Regression - on the dataset. Each model was assessed using three key metrics: the coefficient of determination (R^2), Mean Squared Error (MSE), and Mean Absolute Error (MAE) expressed as a percentage. The results are summarized for both training and testing datasets to provide insights into each model's generalization capability.

Random Forest Regression

The Random Forest model demonstrated the best overall performance among all the models evaluated. On the training data, it achieved an R^2 score of 0.92, an MSE of 0.37, and a MAE of 8.7%. On the testing dataset, the R^2 score slightly declined to 0.78, with an MSE of 0.80 and a MAE of 10.8%. These results suggest that Random Forest generalizes well and maintains low prediction error, with only a modest increase in error metrics on unseen data.

Decision Tree Regression

There was not much overfitting in the Decision Tree model's performance throughout the training and testing datasets. It obtained an R2 value of 0.77 in testing and 0.81 in training. Training had an MAE of 12.7%, whereas testing had an MAE of 10.8%. The Decision Tree had a good baseline and produced results that were competitive, even though it was a simpler model than ensemble techniques.

Support Vector Regression (SVR)

The SVR model achievement of R^2 score of 0.72 along with MAE of 14% based on training dataset. However, performance declined on the test set, with the R^2 dropping to 0.65 and the MAE increasing to 21%. This suggests a tendency to overfit, as evidenced by the gap in predictive accuracy between training and testing data. While SVR had reasonable training performance, its generalization was weaker relative to ensemble methods.

MLP Regression

The Multilayer Perceptron (MLP) Regression model achieved an R^2 value of 0.70 when working on training data, indicating a moderate level of explained variance. The Mean Squared Error (MSE) of 0.71 and Mean Absolute Error Percentage of 11% suggest that the model fits the training data reasonably but with notable prediction error. On the testing dataset, the model recorded a slightly higher R^2 score of 0.70, which implies comparable generalization performance. However, the increase in MSE to 1.1 and Mean Absolute Error Percentage to 18% indicates that the model is less precise on unseen data.

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XGB Regression

The XGBoost Regressor demonstrated strong predictive performance across both evaluation phases. During training, it achieved an R^2 score of 0.88, a Mean Squared Error (MSE) of 0.56, and a Mean Absolute Error (MAE) of 10.5%, indicating a high level of fit to the training data. On the test set, the model maintained generalization capability with an R^2 score of 0.77, an MSE of 0.86, and an MAE of 12%. These results reflect the model's effectiveness in capturing complex patterns and maintaining reasonable error margins.

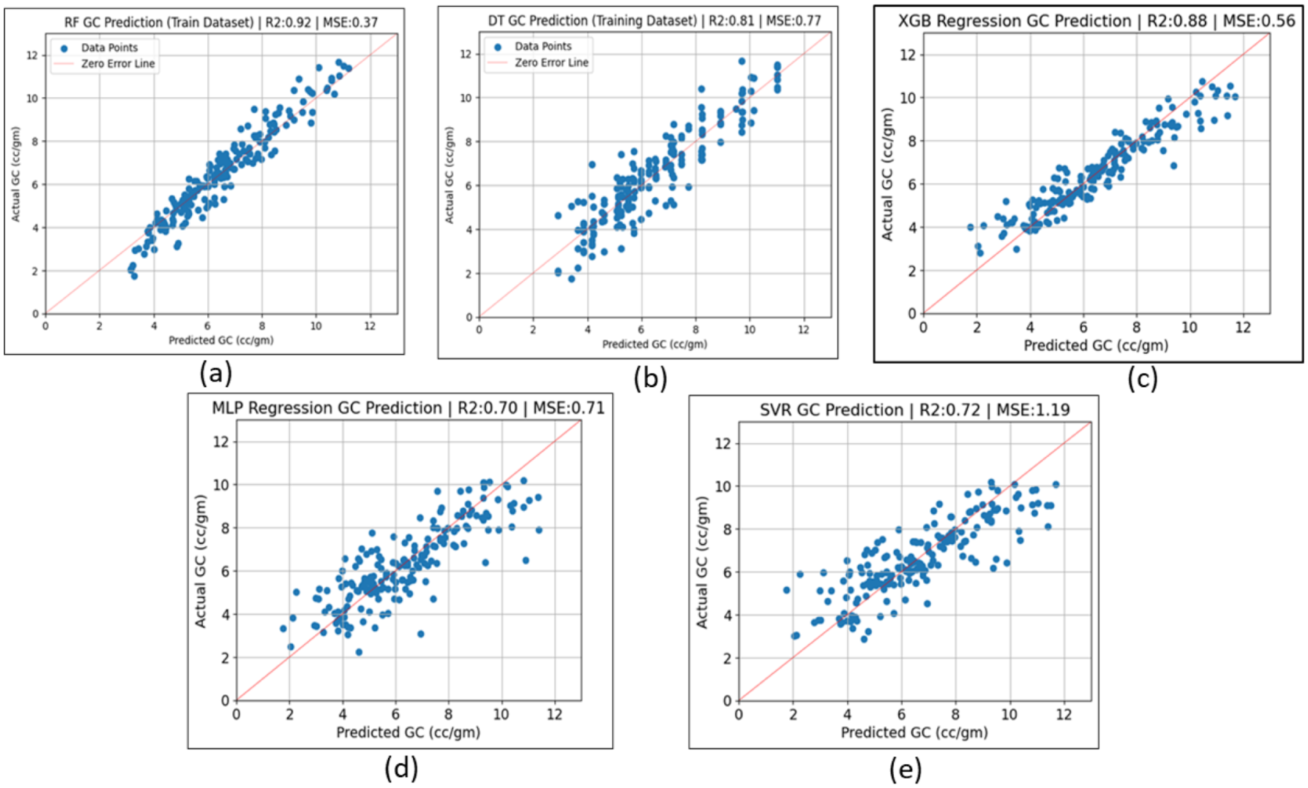

Figure 15—*Gas Content Prediction in Training Dataset 15(a) Random Forest; 15(b) Decision Tree; 15(c) XGB regression; 15(d) MLP Regression; 15(e) SVM regressor (SVR)*

Among the evaluated models, Random Forest and XG-Boost regression emerged as the most effective in terms of predictive accuracy and generalization. While the MLP and Support Vector Regression (SVR) models were included for their capability to model complex, non-linear relationships, their performance on this dataset did not meet expectations. The Decision Tree model maintained a good balance between simplicity and interpretability but fell short in overall accuracy. Based on a thorough comparative analysis of multiple regression models, including evaluation of R^2 score, Mean Squared Error (MSE), and Mean Absolute Error (MAE), the Random Forest Regressor was identified as the most effective model. Although XGBoost Regression also demonstrated good results, it was ultimately excluded due to its relatively higher MSE and MAE (%) compared to Random Forest. This model was subsequently used for all further analysis and predictive tasks.

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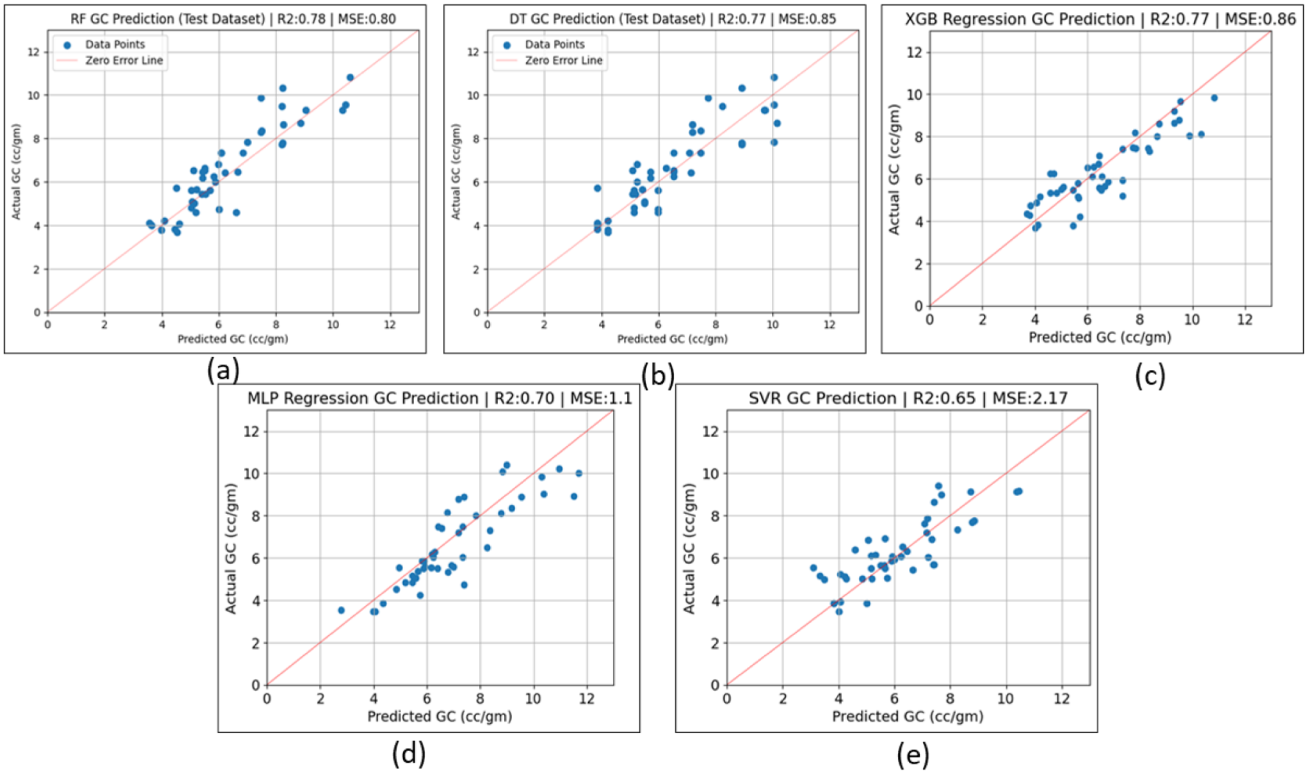

Figure 16—*Gas Content Prediction in Testing Dataset 16(a) Random Forest 16(b) Decision Tree 16(c) XGB regression 16(d) MLP Regression 16(e) SVM Regressor (SVR)*

Validation of Results

For further validation and forecasting part of the total study area selected and the Random Forest model was applied to five blind core wells distributed in the identified area. It is to be mentioned data of these five core wells were not used during training or testing. The model's predictive performance on these wells was evaluated using the R^2 score, which ranged between 80% and 95%, indicating strong generalization capability and reliable matching of 87% on unseen data as represented below Fig.-17:

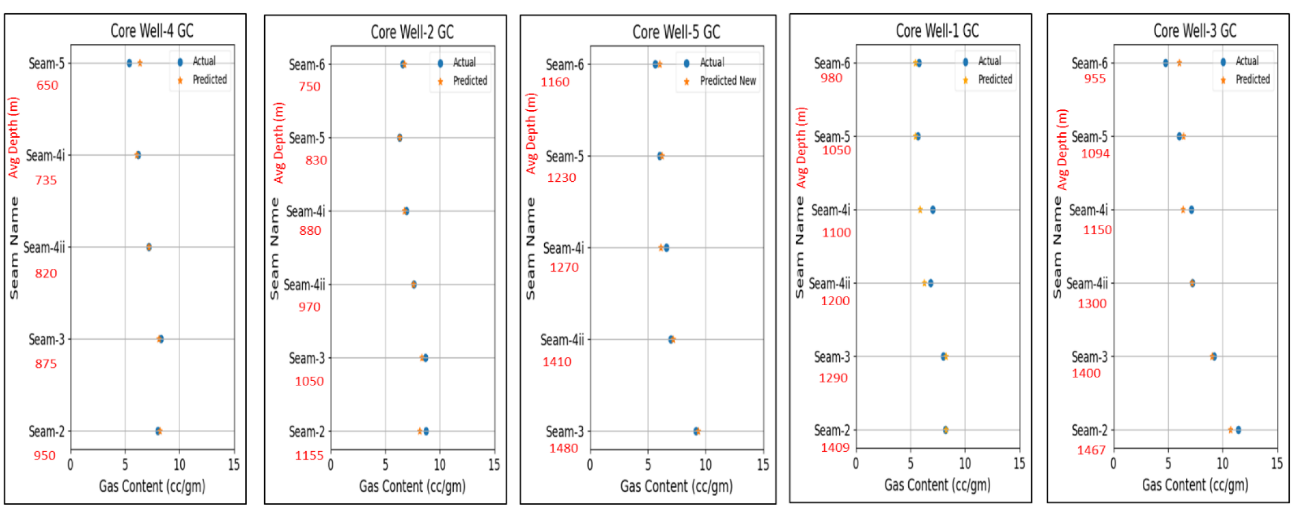

Figure 17—*Gas Content Prediction for Validation in Blind Core Wells*

Table-4 shows the actual and predicted gas content, while Table 5 presents the validation metrics.

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Table 4—Gas Content Prediction Results in Blind Core Wells for Validation

Seam	Core Well - 1		Core Well - 2		Core Well - 3		Core Well - 4		Core Well - 5	
	Predicted GC	Actual GC	Predicted GC	Actual GC	Predicted GC	Actual GC	Predicted GC	Actual GC	Predicted GC	Actual GC
Seam-6	5.43	5.72	6.70	6.60	6.01	4.77	-	-	6.03	5.64
Seam-5	5.43	5.60	6.30	6.30	6.38	6.03	6.36	5.42	6.28	6.04
Seam-4i	5.87	7.02	6.70	6.95	6.34	7.12	6.06	6.20	6.16	6.62
Seam-4ii	6.25	6.80	7.50	7.63	7.16	7.21	7.16	7.22	7.14	7.08
Seam-3	8.24	8.01	8.32	8.67	9.03	9.21	8.08	8.37	9.35	9.18
Seam-2	8.23	8.23	8.11	8.75	10.72	11.42	8.20	8.04	-	-

Table 5—Validation Core Wells Performance Metrics

Core Well	R^2 Score	MSE	Mean Absolute Error (%)
Core Well-1	0.78	0.30	7
Core Well-2	0.82	0.10	3
Core Well-3	0.84	0.46	7
Core Well-4	0.96	0.02	1
Core Well-5	0.95	0.08	3

The model showed strong matching between predicted and actual gas content across all wells, with matching percentages corresponding to R^2 scores ranging from 78% to 96% and average matching greater than 85%. Notably, the model achieved its highest performance on Well-4, with an R^2 of 0.96, an exceptionally low MSE of 0.02, and a MAE of only 1%. Similarly, Well-5 showed excellent accuracy with an R^2 of 0.95 and MAE of 3%. Moderate performance was observed for Well-1 and Well-3, where R^2 values were 0.78 and 0.84, respectively, and MAE remained at 7%. Well-2 exhibited both high accuracy and precision, with an R^2 of 0.82 and the lowest MAE among all wells (3%), aside from Well-4. These results demonstrate that the model retains high predictive accuracy across various well datasets, including those it has not seen during training. The consistently high R^2 scores and low error values affirm the model's ability to generalize well to different field conditions and well profiles.

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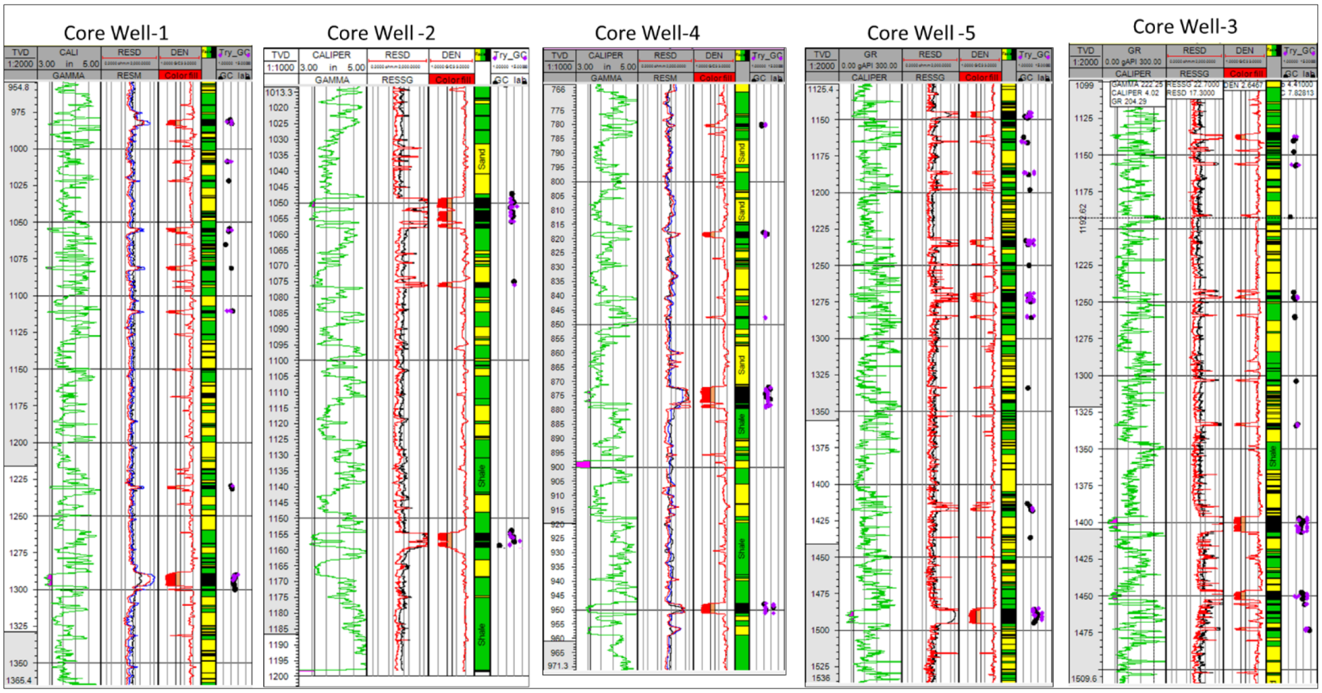

Figure 18—*Gas Content Prediction in Blind Core Wells from Geophysical Logs*

These results further reinforce the robustness and effectiveness of the selected model.

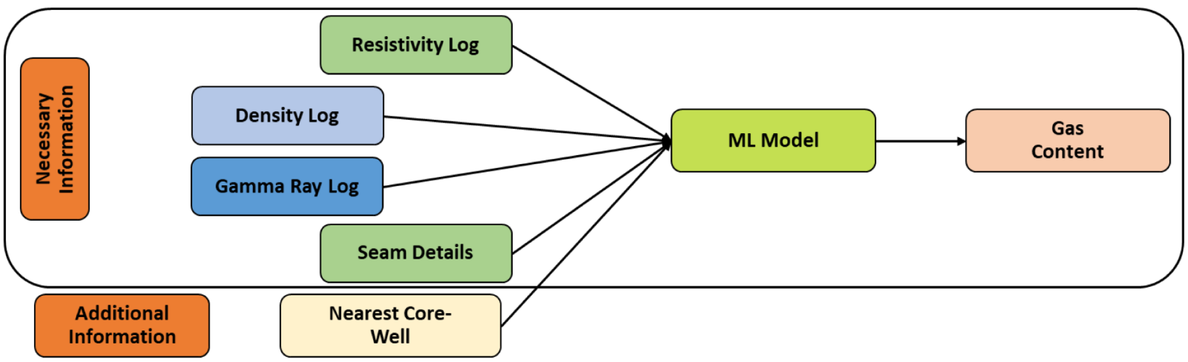

Figure 19—*Prediction from Geophysical Logs Workflow in Development Wells*

Prediction in Development Wells

The Random Forest model was applied to development wells using geophysical logs data and seam information from six wells to predict Gas Content (GC). The predicted GC values showed a strong alignment with measurements from the nearest core wells, achieving a mean absolute error percentage below 10%. This high level of accuracy indicates that the model effectively captures the complex, non-linear relationship between well log responses and gas content. Furthermore, performance metrics such as a high R^2 value and low Mean Absolute Error (MAE) in blind validation wells confirm the model's robustness and predictive reliability. These results demonstrate the model's potential as a valuable tool for estimating GC in wells lacking direct core desorption data. The predicted gas content of development wells and the GC difference (%) from the nearest core well closely match the core well data, as illustrated in Fig-20. Variation in predicted gas content with variation in geophysical logs response in development wells. Seam with good geophysical properties gives good Gas Content as seen in Fig-21.

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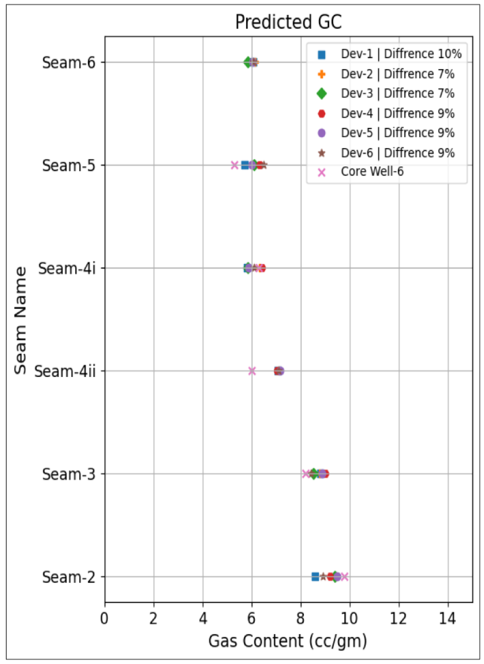

Figure 20—*Prediction Result in Development Wells and Difference in Gas Content from Nearest Core Well*

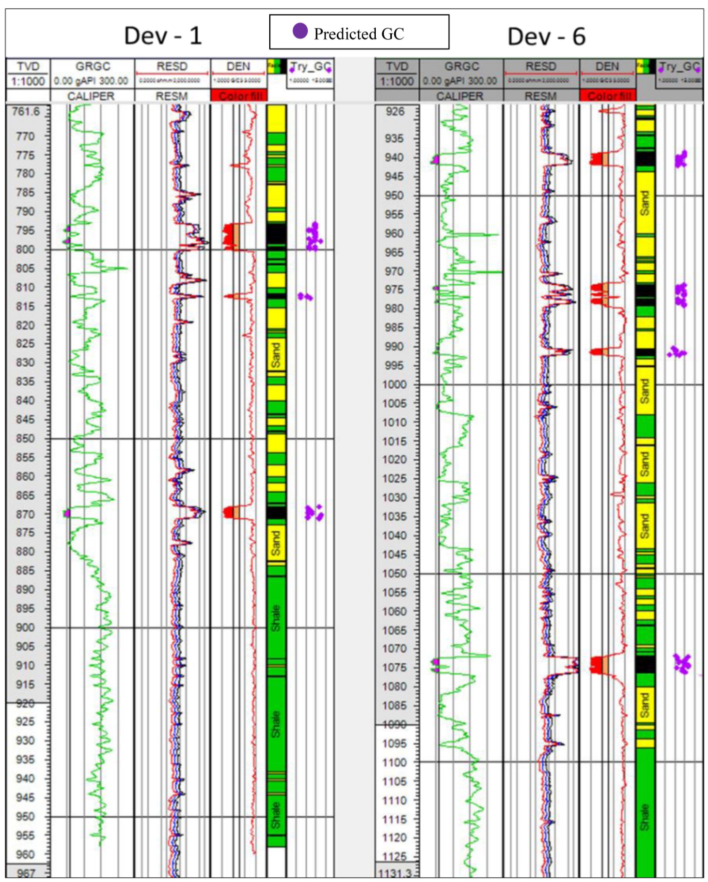

Figure 21—*Variation in Gas Content with Geophysical Logs in Development Wells*

Novelty and Additional Insights

Gas content and resource estimation is one of the key decision point for strategizing the field development plan. The classical method of gas estimation from core wells is time taking and costly affair. Often the

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operators have to plan for a detailed campaign during exploration days or intermittently by distributing the development campaigns in staggered manner. In addition, the heterogeneity of coal properties often too extreme to be covered by core well distribution. This paper showcases it is possible to predict the gas content form geophysical logs and propagate step by step; as the development continues and a significant scale of accuracy may be achieved leveraging the vast data through Machine learning solutions. Utilizing such methods, it is possible to minimize exploration and development costs by reducing uncertainty. A better potential identification shall enable more effective decision making and optimize development plan. As a whole such data driven Machine Learning solutions can play pivotal role in success of any Coal Bed Methane project by eliminating low producers significantly.

Conclusion

This study introduces a machine learning-based approach as a complementary tool to traditional gas content (GC) estimation methods in coalbed methane (CBM) reservoirs. While core desorption remains the standard technique, the proposed method offers a practical, data-driven alternative for predicting GC using geophysical log data-particularly useful in development wells where core data is unavailable. Due to non-linear nature of the relationship between GC and geophysical logs, various machine learning models were tested, with the Random Forest model achieving the highest accuracy. It performed well on both training and testing datasets and showed strong agreement in blind validation against core well data. In development wells, the model successfully captured variations in gas content corresponding to differences in geophysical log responses. These findings highlight the model's reliability and its potential to enhance CBM development.

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